Distillation with No Labels (DINO)

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Outline

- Why DINO exists
- Labeling bottleneck in defense
- Core concepts (teacher–student, multi-crop, loss)
- Preventing collapse
- DINOv2 evolution
- Performance comparisons

The labeling bottleneck in defense

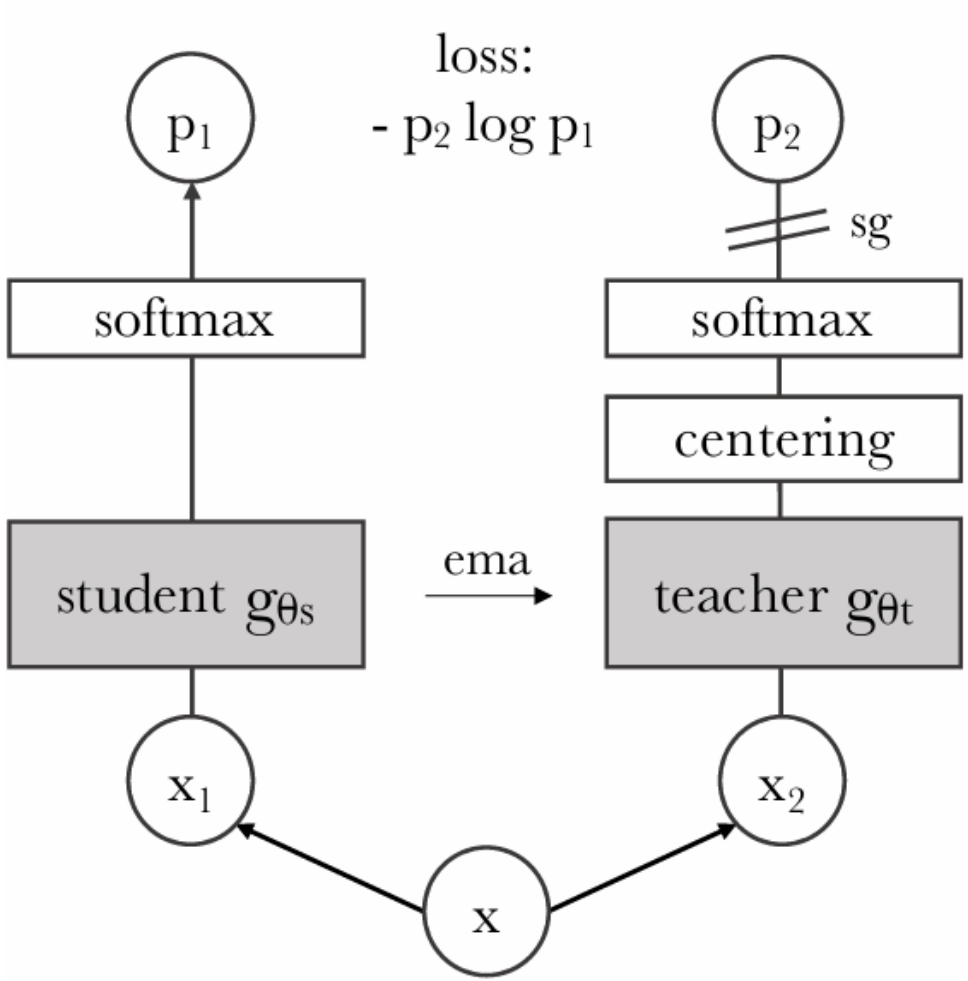
- Labels are expensive and slow to obtain at scale
- Models must transfer across tasks and sensors with minimal retraining
- On-edge constraints favor robuts generalization

DINO in one slide

- DINO - Distillation with NO labels
- A self-supervised distillation method requiring no labels (2021)
- Uses Vision Transformers
- Teacher–student framework
- No contrastive negatives required
- Learns transferable features from unlabeled images

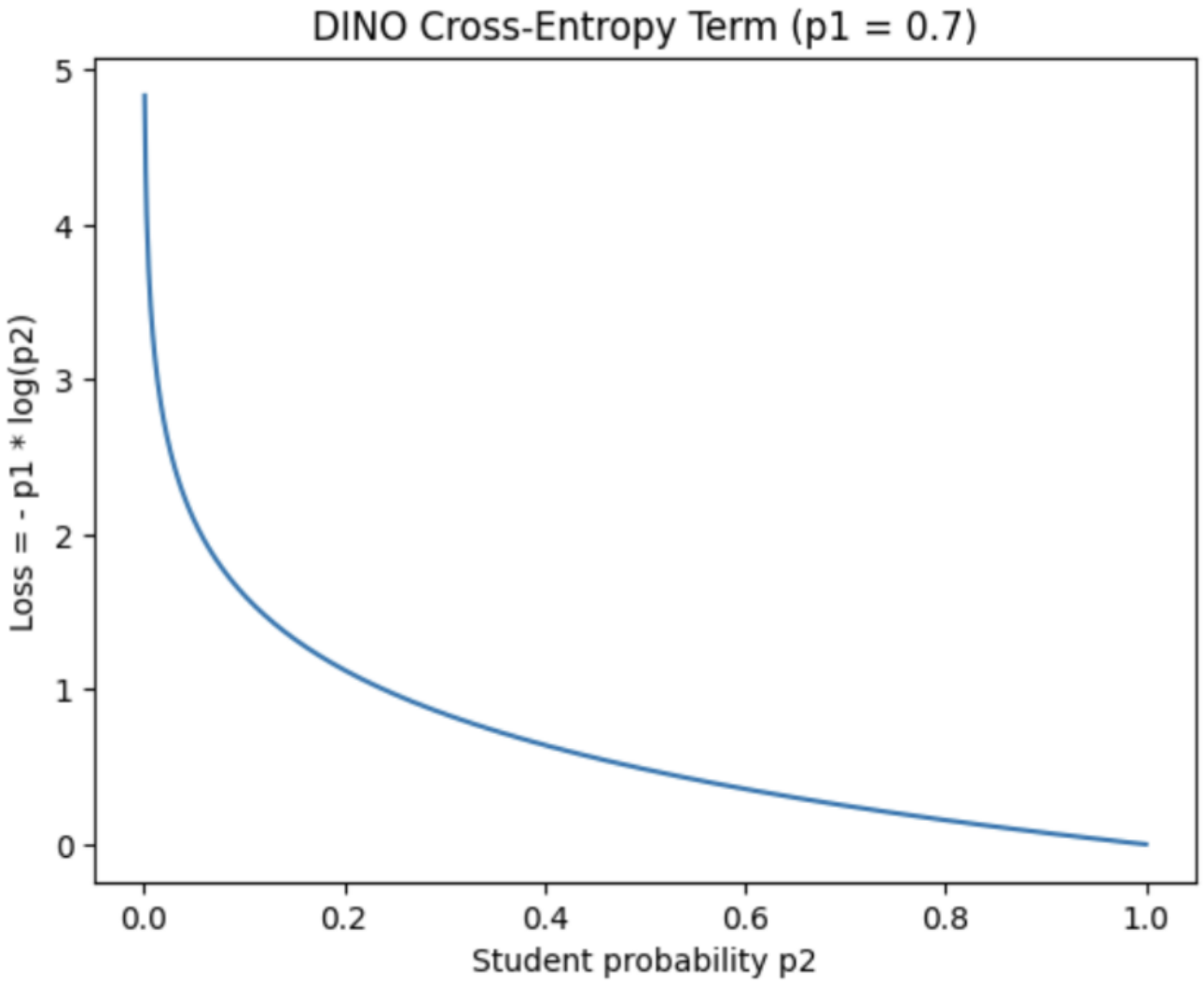
Teacher-Student: how learning happens

- Knowledge Distillation – learning paradigm where the student is taught to match the output of the teacher
- Student network matches teacher's outputs on different views of the same image
- In co-distillation, both networks share the same architecture and learn by matching each other's outputs
- Teacher is dynamically built during training
    - Exponential moving average (EMA) updates slowly, creating a stable teacher that filters out noisy gradient updates
- No Labels are used



Visualization of Loss Function

- The teacher produces sharpened probabilities (low temperature).
- When student probability approaches teacher confidence (high) → loss becomes small.



Multi-Crop Strategy

- Multi-crop teaches the models to connect parts to the whole
- Each image has: x_1^g and x_2^g
  - 2 global crops (224x224)
  - Several local crops (96x96) for the student
- Student sees all crops (global and local)
- Teacher only sees global crops
- Encourages "local-to-global" correspondences

Objective: student matches teacher

- Student matches teacher's probability distribution output
- Minimize cross entropy loss

Without multi-crop

$$\min_{\theta_s} H(P_t(x), P_s(x))$$

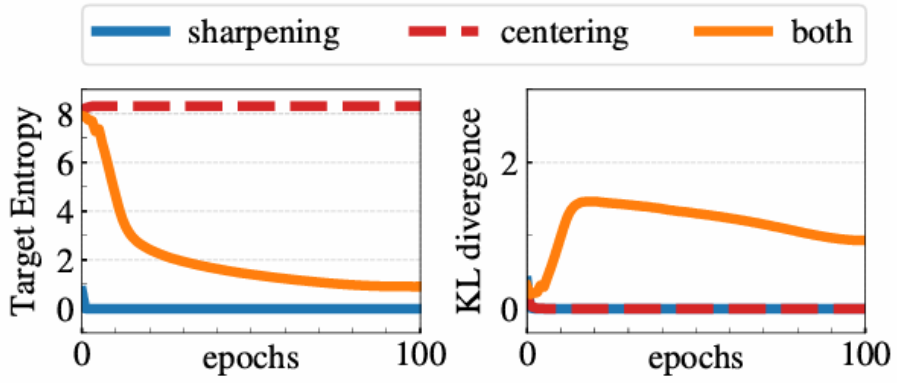
where $H(a, b) = -a \log b$.

Multi-Crop

$$\min_{\theta_s} \sum_{x \in \{x_1^g, x_2^g\}} \sum_{\substack{x' \in V \\ x' \neq x}} H(P_t(x), P_s(x'))$$

Preventing Collapse: why DINO stays stable

- Collapse – when all images map to the same representation
- Centering and sharpening the teacher output balance out to avoid collapse
  - Centering on teacher output for normalization
    - Subtract running mean to avoid single-dimension dominance
  - Sharpening (Temperature Scaling)
    - Teacher outputs use a lower temperature for more peaked softmax probability output
    - Forces confident, informative targets for the student match



Architecture

Backbone: Vision Transformer (patch size of 8 or 16)

Projection head – 3-layer multi-layer perceptron (MLP)

Softmax output to get prob dist

EMA teacher update

Momentum encoder

Emergent Property: Semantic Segmentation

- Self-attention maps encode object boundaries
- Different attention heads specialize for different semantic parts

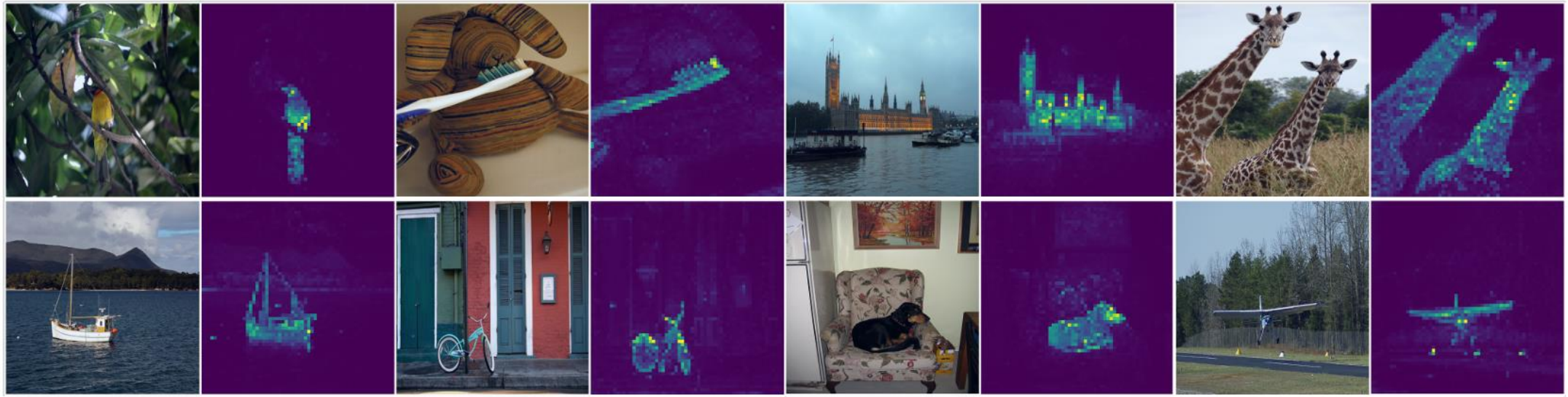


Figure 1: **Self-attention from a Vision Transformer with 8×8 patches trained with no supervision.** We look at the self-attention of the `[CLS]` token on the heads of the last layer. This token is not attached to any label nor supervision. These maps show that the model automatically learns class-specific features leading to unsupervised object segmentations.

DINOv2

- 1B-parameter ViT trained on curated 142M-image dataset (LVD-142M)
  - Automatically curated using retrieval pipeline
- Combined DINO + iBOT objectives, adding patch-level masked image modeling
- Features work without finetuning
- Better general features from better data + objective blend

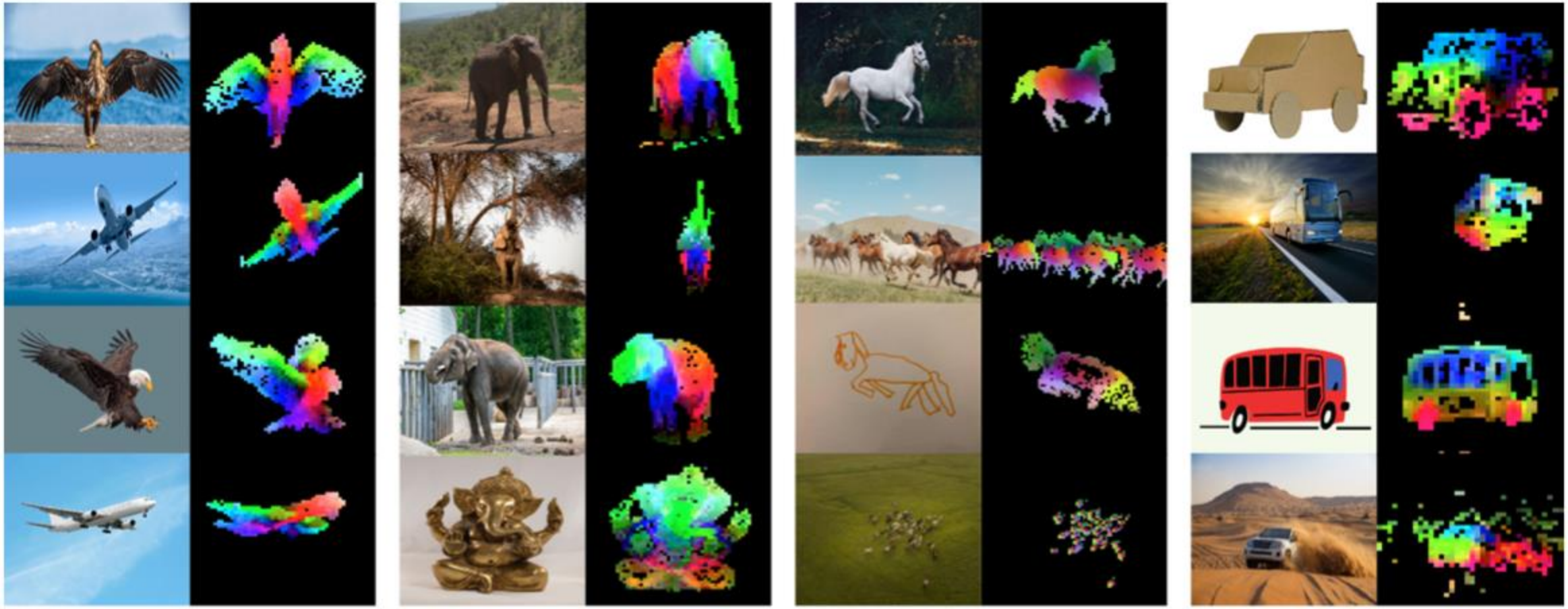


Figure 1: **Visualization of the first PCA components.** We compute a PCA between the patches of the images from the same column (a, b, c and d) and show their first 3 components. Each component is matched to a different color channel. Same parts are matched between related images despite changes of pose, style or even objects. Background is removed by thresholding the first PCA component.

Performance Results – Linear and kNN classification

Method	Arch.	Param.	im/s	Linear	k-NN
Supervised	RN50	23	1237	79.3	79.3
SCLR [12]	RN50	23	1237	69.1	60.7
MoCov2 [15]	RN50	23	1237	71.1	61.9
InfoMin [67]	RN50	23	1237	73.0	65.3
BarlowT [81]	RN50	23	1237	73.2	66.0
OBoW [27]	RN50	23	1237	73.8	61.9
BYOL [30]	RN50	23	1237	74.4	64.8
DCv2 [10]	RN50	23	1237	75.2	67.1
SwAV [10]	RN50	23	1237	**75.3**	65.7
DINO	RN50	23	1237	**75.3**	**67.5**
Supervised	ViT-S	21	1007	79.8	79.8
BYOL* [30]	ViT-S	21	1007	71.4	66.6
MoCov2* [15]	ViT-S	21	1007	72.7	64.4
SwAV* [10]	ViT-S	21	1007	73.5	66.3
DINO	ViT-S	21	1007	**77.0**	**74.5**
Comparison across architectures					
SCLR [12]	RN50w4	375	117	76.8	69.3
SwAV [10]	RN50w2	93	384	77.3	67.3
BYOL [30]	RN50w2	93	384	77.4	–
DINO	ViT-B/16	85	312	78.2	76.1
SwAV [10]	RN50w5	586	76	78.5	67.1
BYOL [30]	RN50w4	375	117	78.6	–
BYOL [30]	RN200w2	250	123	79.6	73.9
DINO	ViT-S/8	21	180	79.7	**78.3**
SCLRv2 [13]	RN152w3+SK	794	46	79.8	73.1
DINO	ViT-B/8	85	63	**80.1**	77.4

Performance Results – Transfer Learning

Table 6: **Transfer learning by finetuning pretrained models on different datasets.** We report top-1 accuracy. Self-supervised pretraining with DINO transfers better than supervised pretraining.

	Cifar_{10}	Cifar_{100}	INat_{18}	INat_{19}	Flwrs	Cars	INet
ViT-S/16							
Sup. [69]	**99.0**	89.5	70.7	76.6	98.2	92.1	79.9
DINO	**99.0**	**90.5**	**72.0**	**78.2**	**98.5**	**93.0**	**81.5**
ViT-B/16							
Sup. [69]	99.0	90.8	**73.2**	77.7	98.4	92.1	81.8
DINO	**99.1**	**91.7**	72.6	**78.6**	**98.8**	**93.0**	**82.8**

Performance: steelman vs alternatives

- Contrastive (SimCLR/MoCO): strong with negatives; harder batching
- MIM (MAE/iBOT): efficient; may need more finetuning
- CLIP: great zero-shot with text; limited for non-captioned sensors
- Supervised: best with many labels; brittle under domain shifts

- If labels/captions are scarce/sensitive, DINO is a strong starting point

Questions?